

CoRoT-2b

R-1086

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_141019 · created 2026-07-30T16:11:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456949.6374 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.018
Transit depth (Rp/Rs)^2	3.72 +/- 0.71 [%]
Orbital Inclination (inc)	87.5 +/- 1.29
Airmass coefficient 1 (a1)	0.971 +/- 0.017
Airmass coefficient 2 (a2)	0.0156 +/- 0.0097
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#3 - [119, 267]
Optimal Aperture	3.22
Optimal Annulus	4.99
Transit Duration (day)	0.0954 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.018 vs 0.1667 (deviation 1.46σ)
Duration fit vs published	0.0954 d vs 0.095 d (deviation 0.11σ)
Mid-time O–C	-3.5 min
Depth SNR	10.7
Residual scatter	1.76 %

Light curve

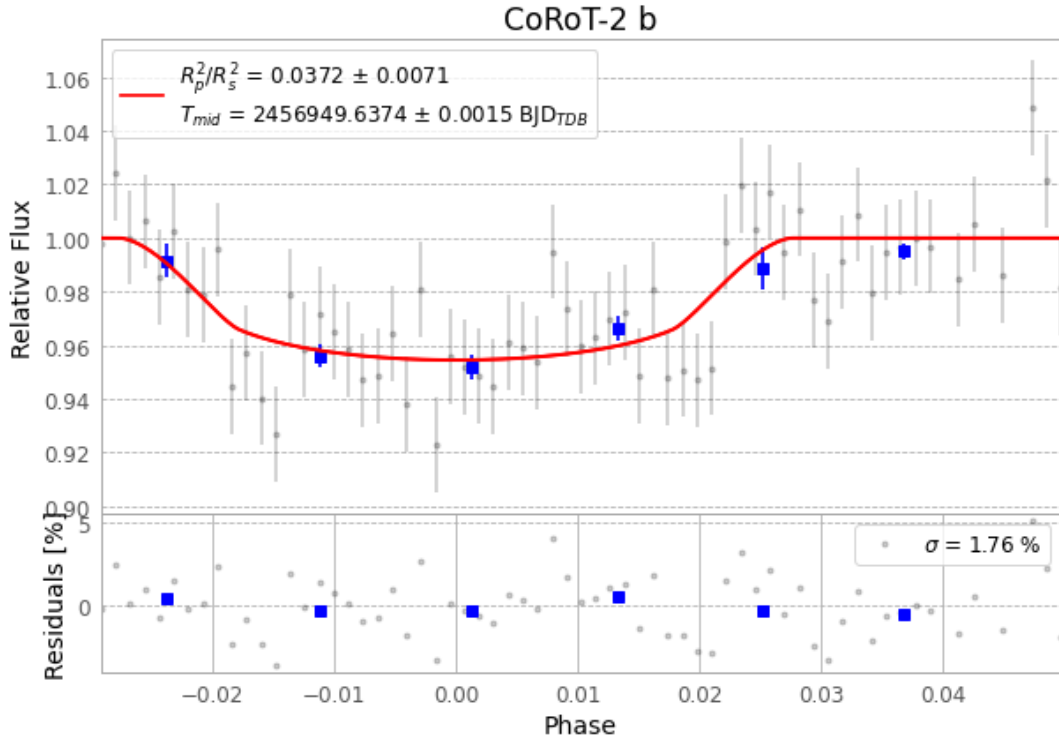


Figure 1 — FinalLightCurve_CoRoT-2 b_2014-10-18.png (SHA-256 b937c657b268b036...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 327], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a1e28d80694b8540...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), CoRoT-2141019020312.FITS → CoRoT-2141019052112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-141019.log (5656f3ebab80...), FinalLightCurve_CoRoT-2 b_2014-10-18.png (b937c657b268...), FinalLightCurve_CoRoT-2 b_2014-10-18.csv (8ef5d6de9374...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 16:11Z.